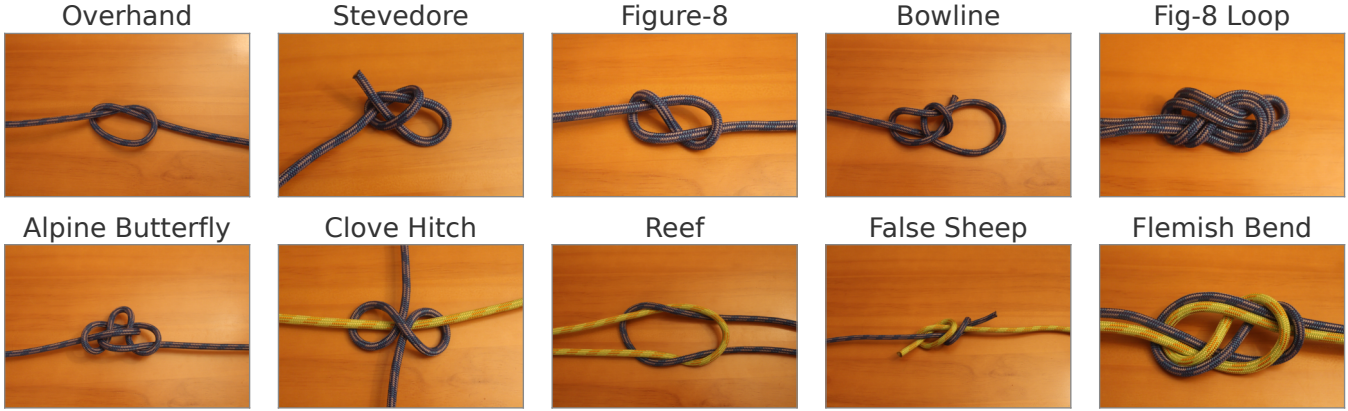


10 Knot Types
3 Tightness Levels
1,440 images

→

CNN Backbone
+ TACA Loss

→

97.2% Acc
(Swin-T)

Key Findings

- Topological distance predicts visual confusion ($p < 0.01$)
- TACA boosts embedding alignment by 40%
- 58–69 pp cross-domain gap from rope appearance bias